Pranav Devara

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	4.35
FIITJEE	Class XII (SSC)	2023	77.50%
Sri Chaitanya	Class X (CBSE)	2021	97.20%

SKILLS

- **Programming Languages:** Python, C, C++, JavaScript, SQL, Shell Scripting, Assembly (RISC-V)
- **AI & Machine Learning:** PyTorch, TensorFlow, Pandas, NumPy, Matplotlib Scikit-learn, SAM2, YOLOv8, scikit-learn, Hugging Face, LangChain data visualisation, SQL, vector embeddings, MLOps basics.
- **Systems & Cloud:** Google Cloud Platform (GCP), AWS (Lambda, DynamoDB), Distributed Systems, gRPC, Node.js, Docker, Git.

RELEVANT COURSES

Computer Science: Data Structures and Algorithms, Computer Architecture, Operating Systems, Database Management(DBMS), System Design, Web Security, Algorithms.

Mathematics & AI: Linear Algebra, Discrete Mathematics, Probability and Statistics.

EXPERIENCE

Data Science & Machine Learning Intern | *Solaiera*

worked in 2 projects Vaidushi & Odentoai

- **Architected a zero-touch batch processing pipeline** for automated dataset curation, rapidly ingesting bulk clinical imagery and converting them into model-ready YOLOv8 annotations and tensor formats using **OpenCV, NumPy**, and multiprocessing techniques.
- **Engineered an automated annotation pipeline (Tag Forge)** leveraging **SAM2** for zero-shot polygon segmentation, replacing high-latency VLMs with **metadata-driven heuristics**.
- **Architected an asynchronous ML inference pipeline** handling concurrent **REST API calls**, decoupling heavy model execution from frontend routing via event-driven message queues to maximize throughput.
- **Optimized LLM token consumption** and generation latency by implementing **semantic request caching** and **context-window compression**, significantly reducing redundant API overhead and compute costs
- **Built Vaidushi, a serverless RAG powered exam generation engine** on AWS (**Lambda, DynamoDB, Pinecone, Gemini LLM**), featuring **async job orchestration** and real-time status tracking, deployed via **SAM**.
- **Optimized Vaidushi's data model with a modular, plug and play architecture**, decoupling frequently used services (Injection, Retrieval, **AWS Cognito** authentication, **Caching module**, **Embedding module** and **LLM fallback module**) into standalone, reusable modules callable across projects.
- **Engineered a system-prompt registry table** with type based categorisation, enabling direct LLM invocation and faster feature additions without code duplication.
- **Redesigned the S3 folder structure** for scalable, organised data storage, and **migrated the chunking strategy to Hierarchical Chunking**, improving retrieval accuracy and context relevance.

PROJECTS

architect4-ai | Node.js, LLM (Llama 3.3 via Groq), Mermaid, Puppeteer

- Built a CLI/web tool that turns a plain-English project description into a production-ready system design: ER diagrams, AWS architecture, REST API spec, and Terraform snippets, generated via LLM and rendered to PDF
- Implemented automated design validation (dangling FKs, missing PKs) and self-healing diagram generation that retries failed Mermaid renders automatically

Dental Arch Labeler (AB) | Python, YOLOv8, OpenCV, PyQt6

- Developed a desktop application using a fine-tuned YOLOv8 model to detect and auto-number teeth in intraoral images, sorting detections by dental arch position for faster clinical annotation

WisdomBERT | Python, PyTorch

- Designed and trained a lightweight transformer encoder from scratch (custom tokenizer, positional encoding, multi-head attention) to classify dental clinical notes into treatment categories, handling noisy/misspelled clinical text with high accuracy

RISC-V Simulator | C

- Built a RISC-V ISA simulator in C from scratch -- custom assembler (lexer/parser/emitter), instruction execution engine with breakpoints and step debugging, plus a configurable cache simulator (variable size/associativity, LRU/FIFO/random replacement, write-back/write-through) with hit/miss statistics

EXTRACURRICULAR

- **Sponsorship Coordinator (Elan & Vision):** Led corporate outreach and strategic partner integrations for IIT Hyderabad's largest tech-cultural festival (35,000+ attendees), successfully managing core promotional assets.
- **Kludge (Cybersecurity Club):** Actively participate in university wide Capture The Flag (CTF) competitions; developing practical skills in network security, binary exploitation, and reverse engineering.